

Hands on activity.<4.4>	
<Character And Strings >	
Course Code: CPE007	Program: Computer Engineering
Course Title: Programming logic and design	Date Performed: 9/24/2025
Section: CPE11S1	Date Submitted:9/25/2025
Name(s): Hanz Rhayven M, Pabalan	Instructor: Engr. Jimlord Quejado

6. Output

character and strings.cpp

```

1  #include <iostream>
2  #include <cctype> // for islower, isupper, toupper, tolower
3
4  using namespace std;
5
6  int main() {
7      cout << "According to islower:\n\n";
8      char lowerCheck[] = {'p', 'P', '5', '!'};
9      int lowerSize = sizeof(lowerCheck)/sizeof(lowerCheck[0]);
10
11     for (int i = 0; i < lowerSize; i++) {
12         char c = lowerCheck[i];
13         if (islower(c))
14             cout << c << " is a lowercase letter\n";
15         else
16             cout << c << " is not a lowercase letter\n";
17     }
18
19     cout << "\nAccording to isupper:\n\n";
20     char upperCheck[] = {'D', 'd', '8', '&'};
21     int upperSize = sizeof(upperCheck)/sizeof(upperCheck[0]);
22
23     for (int i = 0; i < upperSize; i++) {
24         char c = upperCheck[i];
25         if (isupper(c))
26             cout << c << " is an uppercase letter\n";
27         else
28             cout << c << " is not an uppercase letter\n";
29     }
30
31     cout << "\nConversions:\n\n";
32     // Convert to uppercase
33     char toUpper[] = {'u', '7', '$'};
34     int upperConvSize = sizeof(toUpper)/sizeof(toUpper[0]);
35
36     for (int i = 0; i < upperConvSize; i++) {
37         char c = toUpper[i];
38         cout << c << " converted to uppercase is " << (char)toupper(c) << "\n";
39     }
40
41     // Convert to lowercase
42     char toLower[] = {'L'};
43     int lowerConvSize = sizeof(toLower)/sizeof(toLower[0]);
44
45     for (int i = 0; i < lowerConvSize; i++) {
46         char c = toLower[i];
47         cout << c << " converted to lowercase is " << (char)tolower(c) << "\n";
48     }
49
50     return 0;
51 }

```

Output:



C:\Users\hanz\OneDrive\Docu



According to islower:

```
p is a lowercase letter
P is not a lowercase letter
5 is not a lowercase letter
! is not a lowercase letter
```

According to isupper:

```
D is an uppercase letter
d is not an uppercase letter
8 is not an uppercase letter
& is not an uppercase letter
```

Conversions:

```
u converted to uppercase is U
7 converted to uppercase is 7
$ converted to uppercase is $
L converted to lowercase is l
```

```
-----
Process exited after 1.865 seconds with return value 0
Press any key to continue . . . |
```

Explanation:

This program starts by including `<iostream>` for input and output and `<cctype>` so I can use functions like `islower`, `isupper`, `toupper`, and `tolower`. Inside `main()`, the first part checks if certain characters are lowercase. I have an array `lowerCheck` with characters like 'p', 'P', '5', and '!'. Using a loop, each character is tested with `islower(c)`. If it's lowercase, the program prints that it is, otherwise it says it's not.

The second part does the same but for uppercase letters. It stores characters in `upperCheck` like 'D', 'd', '8', and '&'. With another loop, each character is checked using `isupper(c)` to see if it's uppercase, and the program prints the result.

After the checks, the program moves to conversions. First, it demonstrates converting characters to uppercase. An array `toUpper` is created with characters like 'u', '7', and '\$'. In a loop, each character is passed into `toupper(c)`, and the converted result is printed.

Then, the program shows conversion to lowercase. It creates an array `toLower` with characters like 'L'. In the loop, each character is passed into `tolower(c)`, and the lowercase version is displayed.

then end.

7. Supplementary Activity

2)

```
#include <iostream>
#include <cctype>

void print_test(const char* func_name, bool result) {
    std::cout << func_name << ": " << (result ? "Yes" : "No") << std::endl;
}

int main() {
    char input_char;
    std::cout << "Enter a single character to test: ";
    std::cin >> input_char;

    std::cout << "\n-- Analysis for character '" << input_char << "' --" << std::endl;
    print_test("Is Alphanumeric? (isalnum)", isalnum(input_char));
    print_test("Is Alphabetic? (isalpha)", isalpha(input_char));
    print_test("Is Lowercase? (islower)", islower(input_char));
    print_test("Is Uppercase? (isupper)", isupper(input_char));
    print_test("Is a Digit? (isdigit)", isdigit(input_char));
    print_test("Is Hex Digit? (isxdigit)", isxdigit(input_char));
    print_test("Is Control Char? (isctrl)", isctrl(input_char));
    print_test("Is Printable? (isprint)", isprint(input_char));
    print_test("Has Graphics? (isgraph)", isgraph(input_char));
    print_test("Is Whitespace? (isspace)", isspace(input_char));
    print_test("Is Punctuation? (ispunct)", ispunct(input_char));

    std::cout << "-----" << std::endl;
    std::cout << "Converted to uppercase: " << (char)toupper(input_char) << std::endl;
    std::cout << "Converted to lowercase: " << (char)tolower(input_char) << std::endl;

    return 0;
}
```

OUTPUT:

```
C:\Users\hanz\OneDrive\Docu × + v
Enter a single character to test: B

-- Analysis for character 'B' --
Is Alphanumeric? (isalnum): Yes
Is Alphabetic? (isalpha): Yes
Is Lowercase? (islower): No
Is Uppercase? (isupper): Yes
Is a Digit? (isdigit): No
Is Hex Digit? (isxdigit): Yes
Is Control Char? (iscntrl): No
Is Printable? (isprint): Yes
Has Graphics? (isgraph): Yes
Is Whitespace? (isspace): No
Is Punctuation? (ispunct): No
-----
Converted to uppercase: B
Converted to lowercase: b
-----
Process exited after 6.301 seconds with return value 0
Press any key to continue . . . |
```

EXPLANATION:

I learned about `<cctype>` library. At the beginning, it includes the `<iostream>` library for input and output operations and `<cctype>` for character-related functions. The helper function `print_test` is defined to take in the name of a test and a boolean result, then display whether the condition is true (“Yes”) or false (“No”). In the main function, the program first prompts the user to enter a single character and stores it in the variable `input_char`. It then prints a header indicating which character is being analyzed. A series of checks are performed using functions such as `isalnum`, `isalpha`, `islower`, **`isupper`**, **`isdigit`**, **`isxdigit`**, **`iscntrl`**, **`isprint`**, **`isgraph`**, **`isspace`**, and **`ispunct`**, each determining a specific property of the input character—for example, whether it is alphanumeric, a digit, a whitespace character, punctuation, and so on. For each test, the program calls `print_test` to display the result in a consistent format. After running all the checks, the program prints a separator line and then shows the uppercase and lowercase versions of the input character using **`toupper`** and **`tolower`** functions, respectively. Finally, it returns `0` to indicate successful execution. This structured approach allows the program to thoroughly evaluate and present different characteristics of any given single character input.

3)

```

#include <iostream>
#include <string>
#include <cstdlib>
}
int main () {

    std::string s1, s2, s3, s4;
    std::cout<<"Enter 4 integer value"<<std::endl;
    std::cin>> s1>> s2>> s3>> s4;

    int num1 = atoi(s1.c_str());
    int num2 = atoi(s2.c_str());
    int num3 = atoi(s3.c_str());
    int num4 = atoi(s4.c_str());

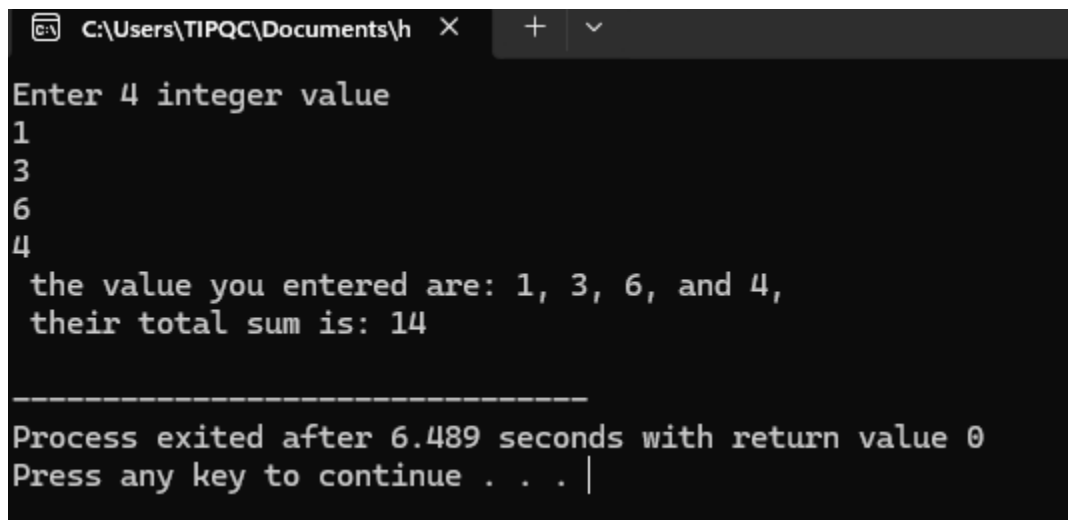
    int sum = num1 + num2 + num3 + num4;

    std::cout<<" the value you entered are: " <<num1<<, " <<num2<<, " <<num3 << ", and " <<num4 <<, " <<std::endl;
    std::cout<<" their total sum is: " <<sum <<std::endl;

    return 0;
}

```

OUTPUT:



```

C:\Users\TIPQC\Documents\h
Enter 4 integer value
1
3
6
4
the value you entered are: 1, 3, 6, and 4,
their total sum is: 14

-----
Process exited after 6.489 seconds with return value 0
Press any key to continue . . . |

```

EXPLANATION:

First, this C++ program includes the required libraries. for handling strings, for input/output, and for conversion functions. To hold user input, four string variables s1, s2, s3, and s4 are declared inside the main. The user is prompted to provide four integer values, which are initially viewed as strings by the computer. After that, each string is transformed into an integer by applying the atoi function to s1.c_str(), s2.c_str(), s3.c_str(), and s4.c_str(), yielding num1, num2, num3, and num4. Following conversion, the software adds up these four integers and records the result in the variable total. After that, it shows each of the entered values individually before printing the sum of them all. Return 0; indicates that the program has successfully completed its execution.

8. Conclusion

This character and string lesson also helped me see how islower() and isupper() work in identifying characters. I now understand that aside from converting text with toupper() and tolower(), I can also check and classify whether a character is uppercase, lowercase, or even a number or symbol. Even if these functions look simple, they give me control over how text is processed, which can really make a big difference when I handle inputs or work with strings in bigger programs.

9. Assessment Rubric